

OSSP – Practical 2

Created the program Practical-2.c and an input file input.txt using nano, then compiled and ran the program. It copied the contents of input.txt into output.txt, and running **cat output.txt** confirms the text was copied successfully.

```
manu@Manasvi:~/os_lab$ nano Practical-2.c
manu@Manasvi:~/os_lab$ nano input.txt
manu@Manasvi:~/os_lab$ gcc Practical-2.c
manu@Manasvi:~/os_lab$ ./a.out
Contents copied successfullymanu@Manasvi:~/os_lab$
manu@Manasvi:~/os_lab$ cat output.txt
This is Practical 2
manu@Manasvi:~/os_lab$ |
```

Used the **strace cat input.txt** command to trace the system calls made by the command. The first part shows the program loading and the shared libraries being mapped into memory.

```

manu@Manasvi::~/os_lab$ strace cat input.txt
execve("/usr/bin/cat", ["cat", "input.txt"], 0x7ffff6c0ac08 /* 26 vars */) = 0
brk(NULL)                               = 0x5c08eb59a000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7a8e09bef000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20887, ...}) = 0
mmap(NULL, 20887, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7a8e09bea000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0-\0\1\0\0\0\0\0\0\0\0\0\0....", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 208288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7a8e09bb7000
mmap(0x7a8e09bb0000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x7a8e09bbe000
mmap(0x7a8e09bbf000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7a8e09bbdf000
mmap(0x7a8e09be0000, 5582, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x7a8e09be6000
mmap(0x7a8e09be8000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7a8e09be8000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0-\0\1\0\0\0\0\0\0\0\0\0\0....", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7a8e09b89000
mmap(0x7a8e09b80000, 174756, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x7a8e09b8d000
mmap(0x7a8e09bb1000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7a8e09bb1000
mmap(0x7a8e09bb5000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x7a8e09bb5000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0-\0\1\0\0\0\0\0\0\0\0\0\0....", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1280152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7a8e09a63000
mmap(0x7a8e09a75000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x7a8e09a75000
mmap(0x7a8e09b15000, 466994, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x7a8e09b15000
mmap(0x7a8e09b87000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x7a8e09b87000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0-\0\1\0\0\0\0\0\0\0\0\0\0....", 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0....", 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0....", 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7a8e09800000
mmap(0x7a8e09828000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7a8e09828000
mmap(0x7a8e099c0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1c0000) = 0x7a8e099c0000
mmap(0x7a8e09a0e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2bd000) = 0x7a8e09a0e000
mmap(0x7a8e09a14000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7a8e09a14000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcr2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0-\0\1\0\0\0\0\0\0\0\0\0\0....", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=599056, ...}) = 0
mmap(NULL, 701112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7a8e09754000
mmap(0x7a8e09757000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x3000) = 0x7a8e09757000
mmap(0x7a8e097d3000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7f000) = 0x7a8e097d3000
mmap(0x7a8e097fe000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xa9000) = 0x7a8e097fe000
close(3)                                = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7a8e09a61000
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7a8e09a5e000
arch_prctl(ARCH_SET_FS, 0x7a8e09a5eb00) = 0
set_tid_address(0x7a8e09a5f128)          = 4025
set_robust_list(0x7a8e09a5ede0, 24)     = 0
rseq(0x7a8e09a5e6a0, 0x20, 0, 0x53053063) = 0
mprotect(0x7a8e09a5e000, 16384, PROT_READ) = 0
mprotect(0x7a8e097fe000, 4096, PROT_READ) = 0
mprotect(0x7a8e09b87000, 4096, PROT_READ) = 0
mprotect(0x7a8e09bb5000, 4096, PROT_READ) = 0
mprotect(0x7a8e09be6000, 4096, PROT_READ) = 0

```

Continued strace output showing further system calls such as openat, read, fstat and close as the kernel sets up and accesses the required files.

```
getrandom("/xd9\xd7\xfa\x8f\xcc\x2\xda\x12", 8, GRND_NONBLOCK) = 0
munmap(0x7a8e09bea000, 20087) = 0
statfs("/sys/fs/selinux", {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffree=0, f_fsid={val=[0x262c91e2, 0x45388667]}, f_namelen=255, f_frsize=4096, f_flags=ST_VALID|ST_NOSUID|ST_NOEXEC|ST_NOATIME}) = 0
statfs("/selinux", 0x7ffe5c255b30) = -1 ENOENT (No such file or directory)
brk(NULL) = 0x5c08eb59a000
brk(0x5c08eb5bb000) = 0x5c08eb5bb000
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbdf....", 1024) = 442
lseek(3, 0, SEEK_CUR) = 442
lseek(3, 416, SEEK_SET) = 416
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.18.33.2-....", 1024) = 1024
read(3, "e 0 0ndevpts /dev/pts devpts rw,....", 1024) = 1024
read(3, "access=client,msize=65536,trans....", 1024) = 1024
read(3, "nr_inodes=1024,mode=700,noswap 0....", 1024) = 556
read(3, "", 1024) = 0
close(3) = 0
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(1, F_GETFD) = 0
fcntl(2, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
poll({fd=0, events=1}, {fd=2, events=1}, 3, 0) = 0 (timout)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7a8e09845cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "5c08abfce000-5c08ac0c000 r--p 0....", 1024) = 1024
read(3, "97fe000-7a8e097ff000 r--p 000a90....", 1024) = 1024
read(3, "00 08:30 13647 /usr/lib/x86_64-lin....", 1024) = 1024
read(3, "0 13641 /us....", 1024) = 566
close(3) = 0
sched_getaffinity(0, 25, [0 1 2 3 4 5 6 7 8 9 10 11]) = 0
rt_sigaction(SIGSEGV, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7a8e09bec000
mprotect(0x7a8e09bec000, 4096, PROT_NONE) = 0
sigaltstack({ss_sp=0x7a8e09bed000, ss_flags=0, ss_size=8192}, NULL) = 0
gettid() = 4025
rt_sigaction(SIGSEGV, {sa_handler=0x5c08ac2f30c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7a8e09845cb0}, NULL, 0) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGBUS, {sa_handler=0x5c08ac2f30c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7a8e09845cb0}, NULL, 0) = 0
prctl(PR_GET_AUXV, 0x7ffe5c255b20, 0x200, 0, 0) = 446
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/uucore", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...}) = 0
read(3, "# Common strings shared across a....", 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7a8e09845cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7a8e09845cb0}, 0) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[RTMIN_RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7a8e09845cb0}, {sa_handler=0x5c08ac2f30c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7a8e09845cb0}, 0) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=[RTMIN_RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7a8e09845cb0}, {sa_handler=0x5c08ac2f30c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7a8e09845cb0}, 0) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|TEXTEN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x5c08eb5dc000) = 0x5c08eb5dc000
```

Final part of the strace output where the file input.txt is opened and read, its contents (This is Practical 2) are printed, and the program exits after closing the file.

```
statx(AT_FDCWD, "input.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=20, ...}) = 0
openat(AT_FDCWD, "input.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 20
splice(4, NULL, 1, NULL, 20, 0) = 1
This is Practical 2
) = 20
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x7a8e09bec000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
manu@Manasvi:~/os_lab$ |
```